Web and Social Media Analysis Towards Recommendations for Improving Business- Shopee

Project Report

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1.0 Introduction

Since its launch in 2015, Shopee Malaysia has grown into a leading e-commerce platform of choice for Southeast Asia. Based in its Kuala Lumpur headquarters, the organization manages a marketplace that accommodates both individual sellers (C2C) and larger brands (B2C). This specific hybrid model depends on an integrated logistics and payment infrastructure to deliver Value to the Customer (V2C). By integrating these systems, the platform effectively simplifies the purchase process and works to reduce the level of transactional friction experienced by users.

As a digital business, Shopee manages Big Data information assets defined by massive traffic volumes and high transactional velocity. Because the platform collects a wide variety of data points from user interactions, it provides a rich environment for performance analysis. This report audits Shopee's digital assets and social sentiment to identify methods for improving Value to the Firm (V2F). This report hence evaluates Shopee's digital assets and social sentiment to provide prescriptive recommendations for long-term growth.

2.0 Web Analytics Landscape

2.1 Web Metrics Comparison

2.1.1 Traffic Volume and Conversion Rate

Shopee Malaysia (shopee.com.my) continues to hold a very strong market presence, as evidenced by the 58.07 million total visits recorded during the month of December 2025. Because of this massive scale of incoming traffic, the platform is officially designated as a "High-Volume" information asset. The actual quality of this visitor engagement is most apparent when looking at the specific "stickiness" found in user behavior. Current records show a notably low Bounce Rate of 24.18%, which effectively means that roughly 76% of all visitors stay to interact with additional content after arriving on their initial landing page (Similarweb, 2025).

These patterns are further supported by an Average Visit Duration of 4 minutes and 57 seconds along with a high average of 7.12 Pages per Visit. Such statistics clearly show that Shopee is successful at leveraging "Intangible Assets" to lengthen the overall customer journey. These

assets specifically include gamification tools like Shopee Games and various live streaming functionalities. This data implies that users are willing to spend significant time on browsing activities. These actions are largely driven by the platform's use of personalized recommendations and various interactive design elements.

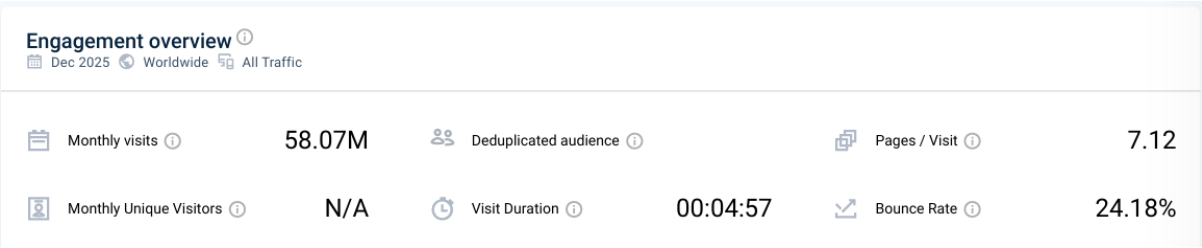


Figure 1**:** *Shopee Malaysia Engagement Overview*

2.1.2 Technical Performance and Load Times

Technical performance is what keeps users from leaving a platform, especially in a mobile-heavy market like Malaysia. Shopee's 2026 Core Web Vitals failure is a major risk for the business. These technical lags create exactly the kind of friction that pushes shoppers to switch to other apps. Industry rules state that any LCP over 4.0 seconds is poor. Shopee has a 6.6 second load time, which means the main content stays hidden for too long. This likely causes micro abandonment where users leave before the page even finishes loading. While the FCP of 2.9 seconds is a bit faster, the entire audit is still a failure. From a business view, these delays are a hurdle for sales during busy events. Shopee needs specific technical fixes to lower these load times if they want to keep their market lead.

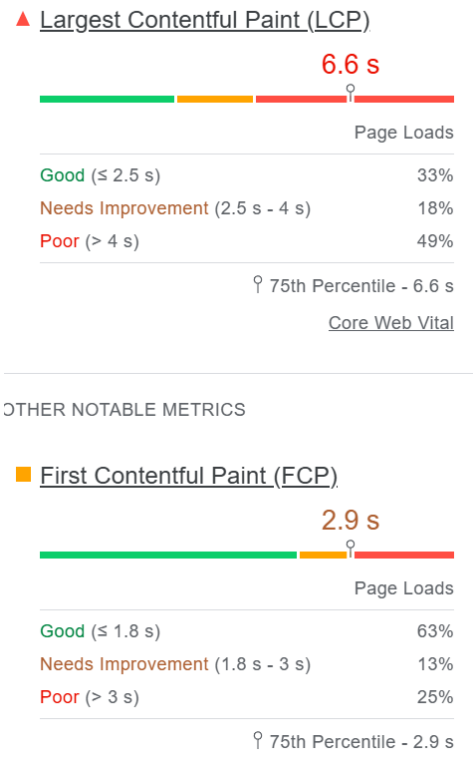


***Figure 2**: Mobile Performance Audit for Shopee Malaysia (Google, 2026)*

In this industry, any LCP that goes over 4.0 seconds is labeled as "Poor." Shopee having a 6.6-second load time means the primary content is hidden for way too long. This situation likely causes micro-abandonment, where shoppers just leave the site before it even finishes rendering. While the FCP is a bit better at 2.9 seconds, the entire audit is still a "Failed" result. From a business standpoint, these delays are a major hurdle for sales during busy shopping periods. Shopee has to make specific technical updates to fix these load times if they want to keep their market lead.

2.1.3 Competitive Context

Shopee is clearly ahead of the competition in Malaysia. It is the 6th most visited website in the country, while its biggest rival Lazada sits much lower at 45th . Data from Similarweb in 2025 shows Lazada has only about 10.3 million visits. This means Shopee gets nearly five times more traffic than its next closest competitorThis massive difference in volume highlights why Shopee is considered the leader in the local e-commerce sector.

There is also the rise of TikTok Shop on the shop.tiktok.com domain to consider. Even though it is a growing name in the social commerce space, that specific domain only recorded 1.5 million visits. This huge gap in traffic numbers makes a very clear distinction between the two

platforms. TikTok is definitely effective at getting people to discover new brands, which represents the concept of Velocity, but Shopee remains the top destination for traffic that is actually ready to complete a transaction, which represents the concept of Volume as per 3 Vs of Big Data.

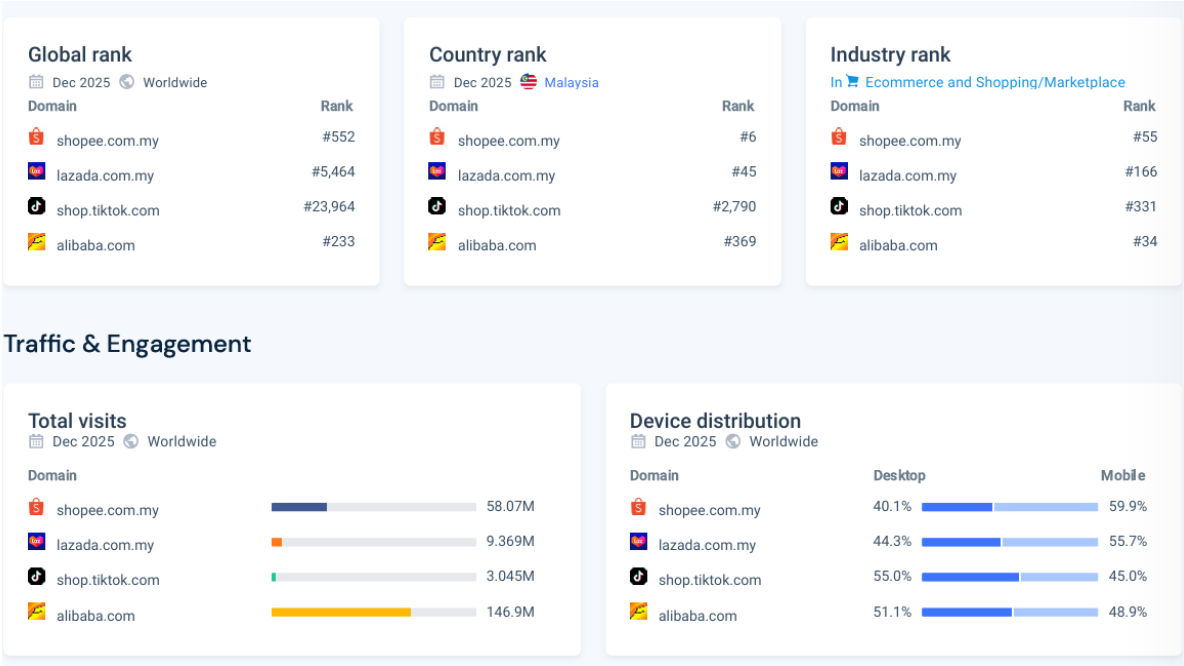


Figure 3: *Comparative Ranking of Shopee vs. Competitors (Similarweb, 2025)*

2.2 Search Engine Optimization (SEO) and Visibility.

2.2.1 Organic Performance and Domain Authority

Organic search contributes 20.83% of total website traffic. This outperforms paid search, which accounts for 12.60%. This organic dominance is driven by a Domain Authority of 89 and a backlink profile containing 3,973 referring domains. These domains serve as high-value digital endorsements that maintain Shopee's high rankings in search engine results pages (SERPs).

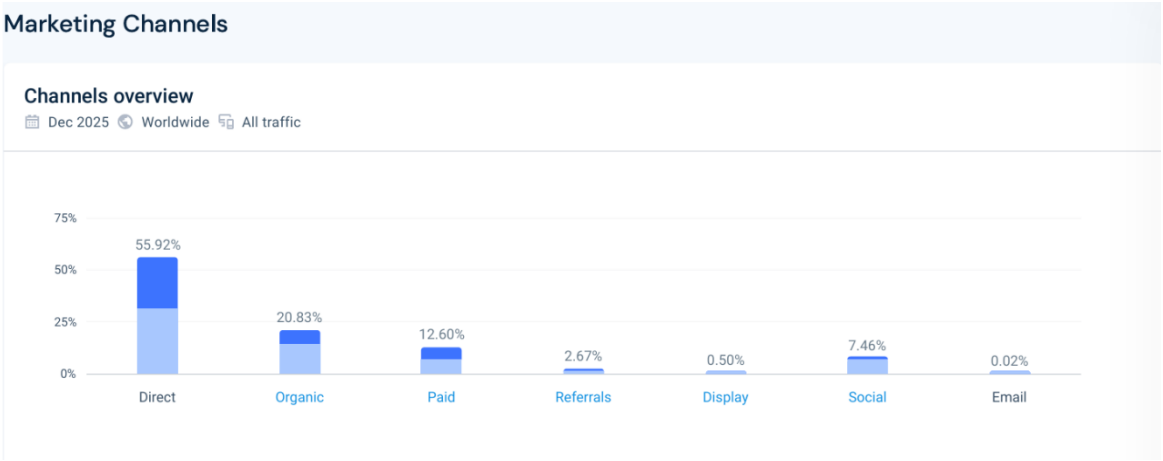


Figure 4: *Traffic Volume and Market Ranking Comparison (Similarweb, 2025)*

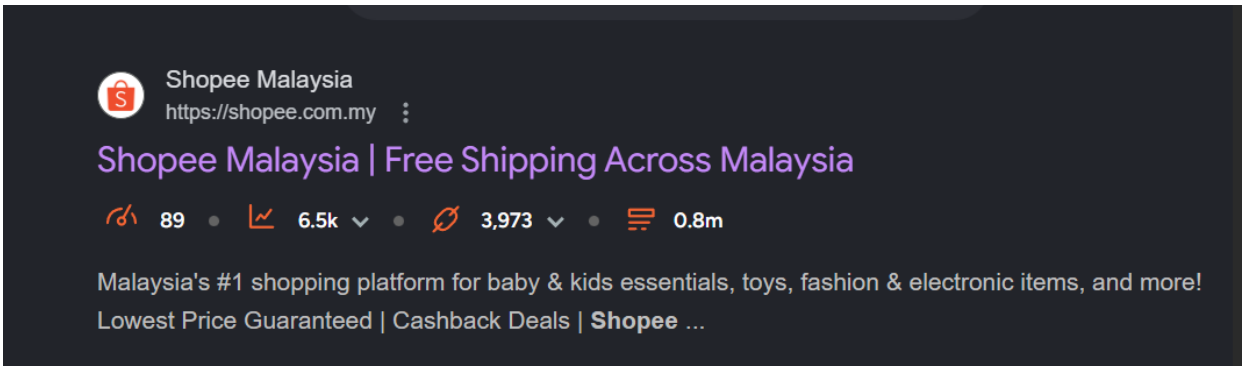
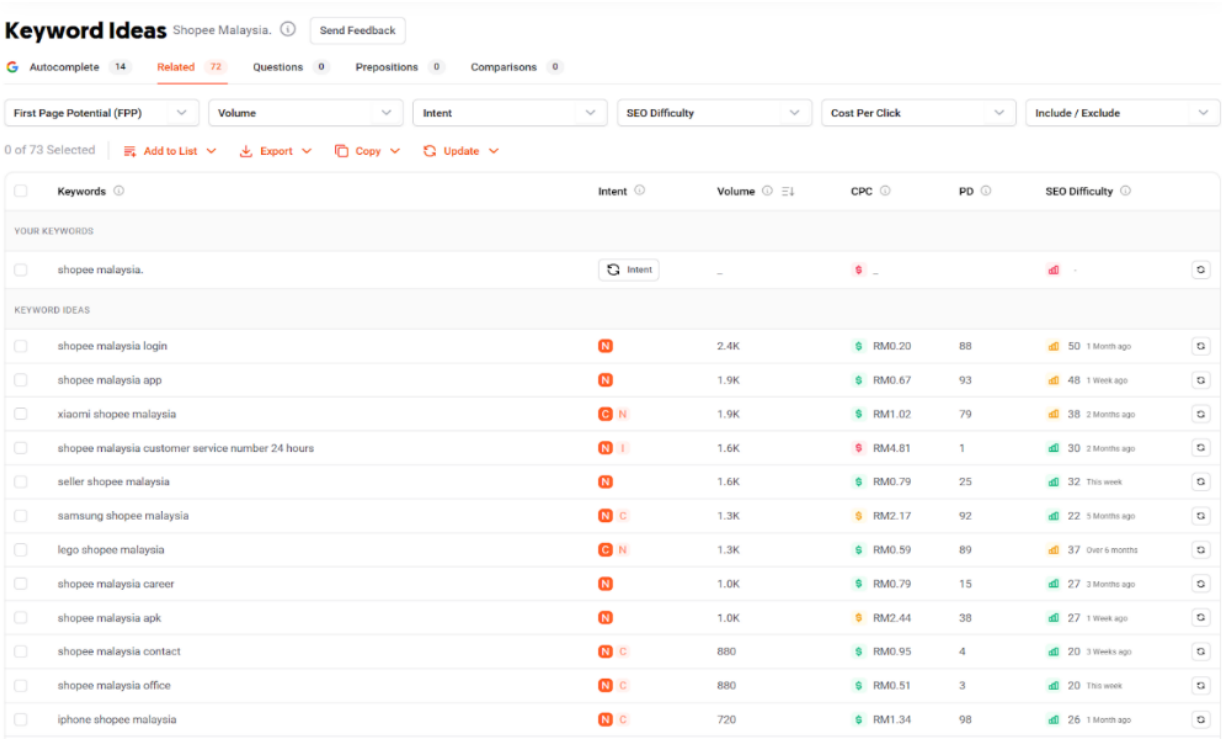


Figure 5: *Search Overview and Domain Authority for Shopee Malaysia (Ubersuggest, 2026).*

If we look at the 2026 Ubersuggest report, Shopee Malaysia has a domain authority of 89. A score like this is the primary reason why their product pages show up at the very top of search results so often. The site is supported by 3,973 referring domains that keep its online presence very strong. It also currently ranks for about 0.8 million organic keywords. Because they have so many keywords, the business can pull in a lot of free traffic instead of always paying for ads. These figures show that Shopee is the top choice for people in Malaysia who use search engines to find products.



Keyword Ideas Shopee Malaysia. Send Feedback

Autocomplete 14 | Related 72 | Questions 0 | Prepositions 0 | Comparisons 0

First Page Potential (FPP) | Volume | Intent | SEO Difficulty | Cost Per Click | Include / Exclude

0 of 73 Selected | Add to List | Export | Copy | Update

Keywords	Intent	Volume	CPC	PD	SEO Difficulty
YOUR KEYWORDS					
shopee malaysia.	Intent	-	-		-
KEYWORD IDEAS					
shopee malaysia login	N	2.4K	RM0.20	88	50 1 Month ago
shopee malaysia app	N	1.9K	RM0.67	93	48 1 Week ago
xiaomi shopee malaysia	C N	1.9K	RM1.02	79	38 2 Months ago
shopee malaysia customer service number 24 hours	N I	1.6K	RM4.81	1	30 2 Months ago
seller shopee malaysia	N	1.6K	RM0.79	25	32 This week
samsung shopee malaysia	N C	1.3K	RM2.17	92	22 5 Months ago
lego shopee malaysia	C N	1.3K	RM0.59	89	37 Over 6 months
shopee malaysia career	N	1.0K	RM0.79	15	27 3 Months ago
shopee malaysia apk	N	1.0K	RM2.44	38	27 1 Week ago
shopee malaysia contact	N C	880	RM0.95	4	20 3 Weeks ago
shopee malaysia office	N C	880	RM0.51	3	20 This week
iphone shopee malaysia	N C	720	RM1.34	98	26 1 Month ago

Figure 6: *Top Organic Keywords and Search Volume Distribution (Ubersuggest, 2026)*

2.2.2 SEO Strategy and Analysis

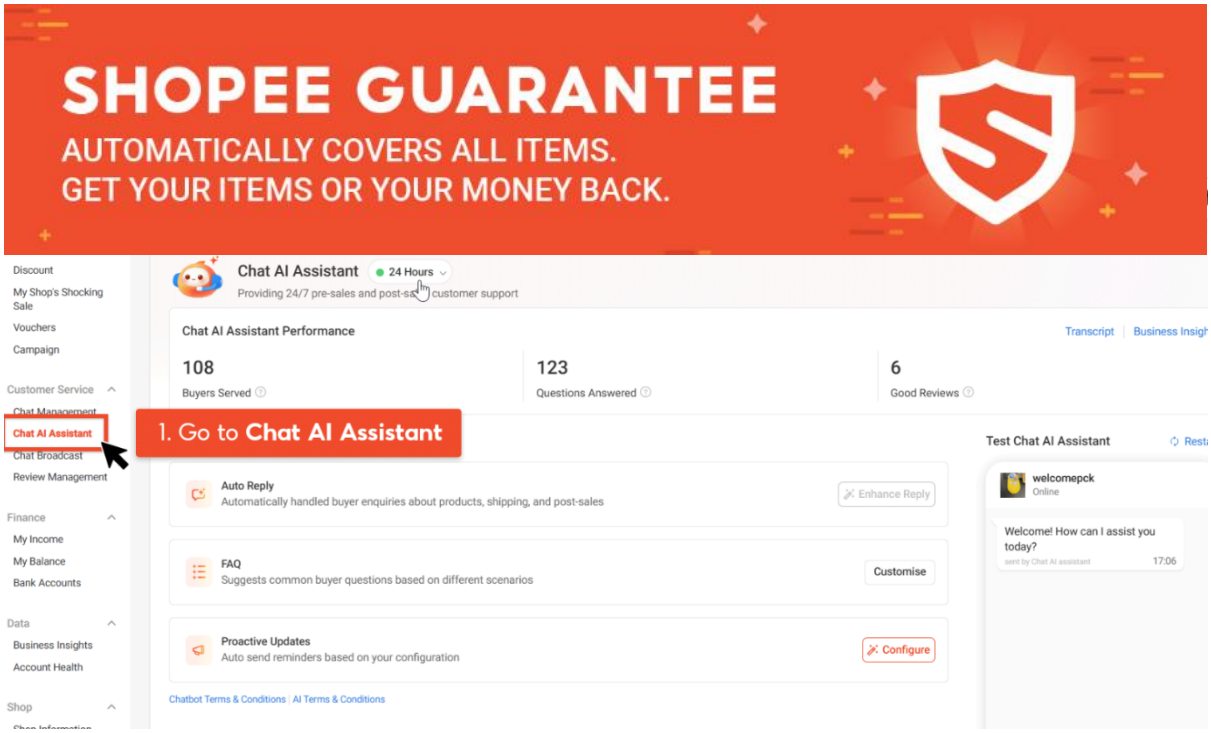
The platform utilizes specific on-page elements to ensure its high-variety data remains discoverable.

SEO Element	Observed Strategy	Strategy Analysis and Business Impact
Keyword Usage	Intent-Based Targeting	Shopee targets 0.8 million keywords. It captures branded traffic (e.g., Shopee login) and non-branded transactional traffic (e.g., Power bank, Baju Kurung).
Meta Descriptions	Conversion Snippets	Descriptions include calls to action like "Buy online" and "Free Shipping." These increase the Click-Through Rate (CTR) from the search results page.
URL Structure	Clean Hierarchy Structure	The site uses clean URLs (e.g., shopee.com.my/product-name). This allows for efficient indexing by crawlers and builds user trust through transparent navigation.

Table 1: *Technical SEO elements*

Generating 20.83% of the 58.07 million total visits through organic search is a primary driver of Value to the Firm (V2F). This high volume of unpaid traffic allows Shopee to suppress its customer acquisition costs, as more than one-fifth of the audience arrives at the platform via non-paid channels. Maintaining such high organic visibility serves as a strategic moat. It ensures the brand remains discoverable in the long term while avoiding the financial necessity of a direct, proportional increase in advertising investment.

2.3 Interactivity and User Engagement



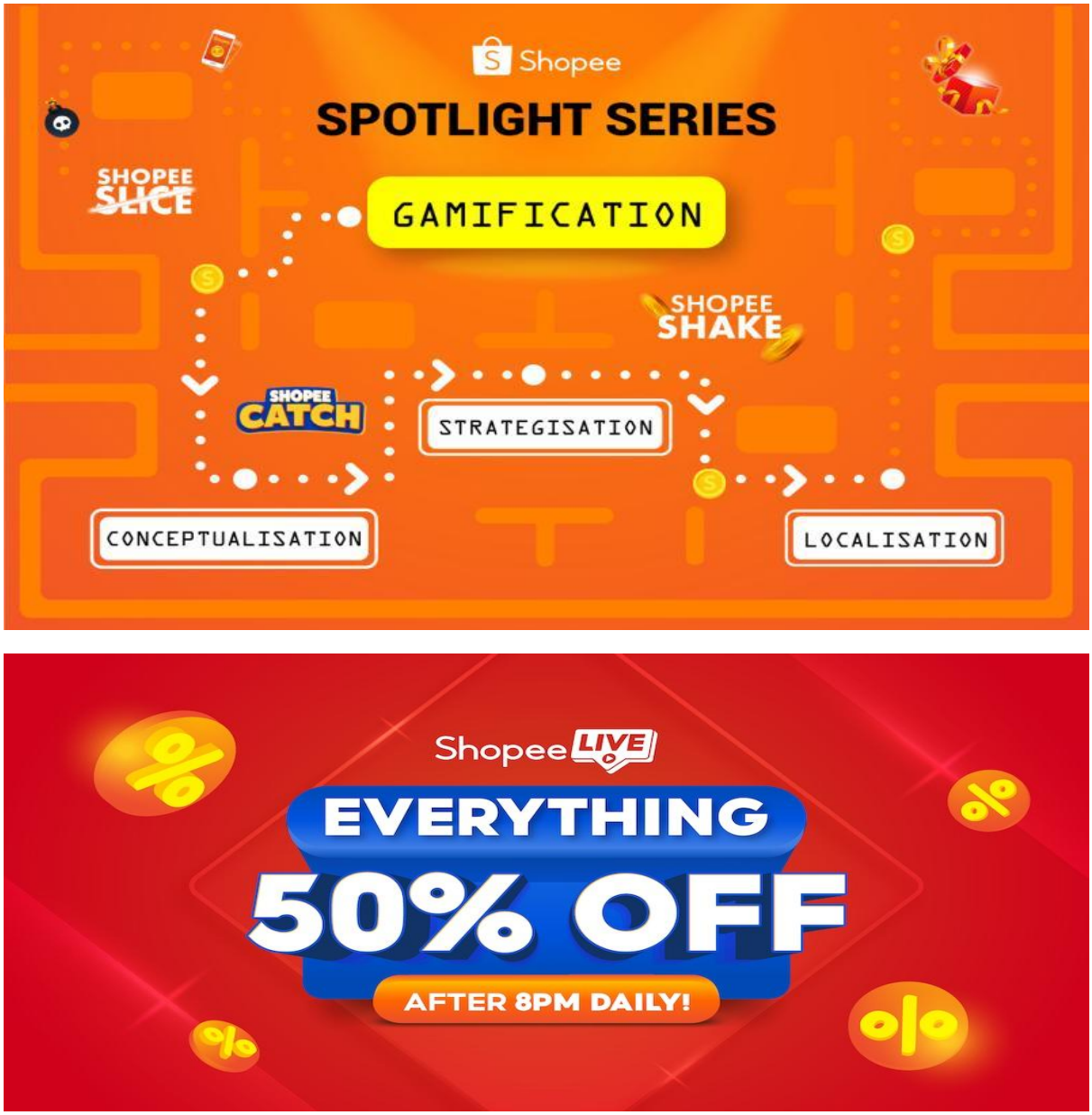


Figure 7: *Social Commerce (AI Chtbot) and Gamification Features*

Figure 8: *Mobile Search Engine Results Page (SERP) Optimization (Google pagespeed, 2026)*

2.3.1 Chat Infrastructure and Support

Users stay on the platform more when they can talk to someone in real time. Shopee uses a two-tier chat system to keep the buying process going. First, the Sophie Chatbot works as an automated first responder and gives people quick answers on promotional vouchers or order tracking. Next is the "Chat with Sellers" tool where buyers talk straight to shop owners (Refer to figure 7). Users can fix issues or talk about prices right inside the app, which means they do not have to leave to find help. This helps Shopee get rid of the friction that often makes people give up and close their tabs. Because of this direct feedback loop, the site is able to keep its bounce rate quite low.

2.3.2 Customer Engagement

Shopee combines Shopee Live and Shopee Coins to move past simple retail and enter social commerce. These features use gamification to encourage repeat visits and grow the time users spend on the platform as shown in figure 7.

2.4 Mobile Optimization

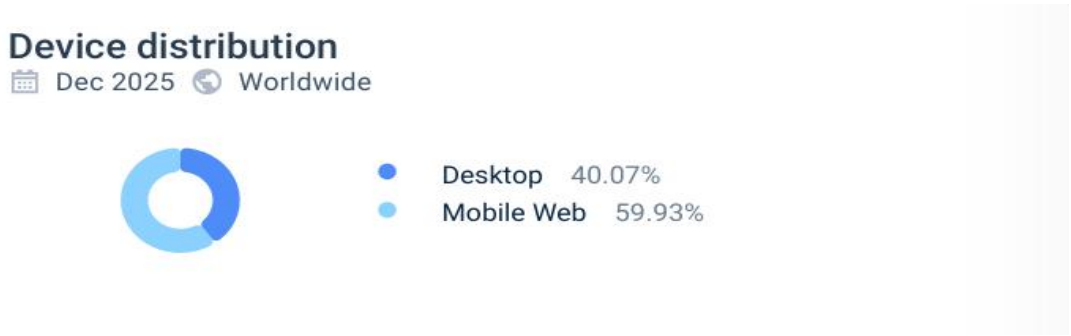


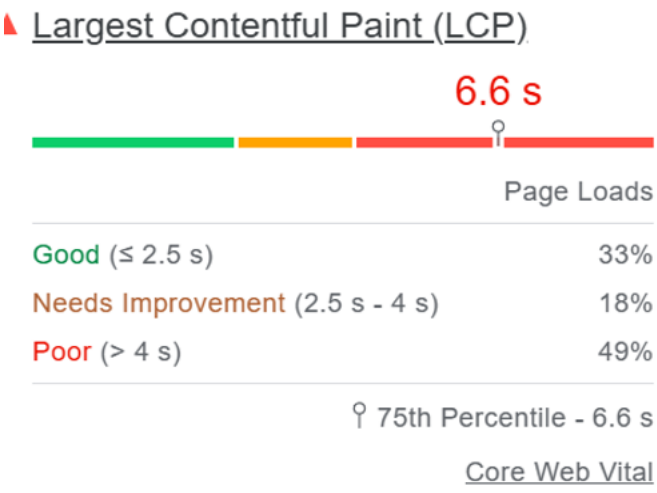
Figure 9: *Mobile vs. Desktop User Distribution (SimilarWeb, 2026)*

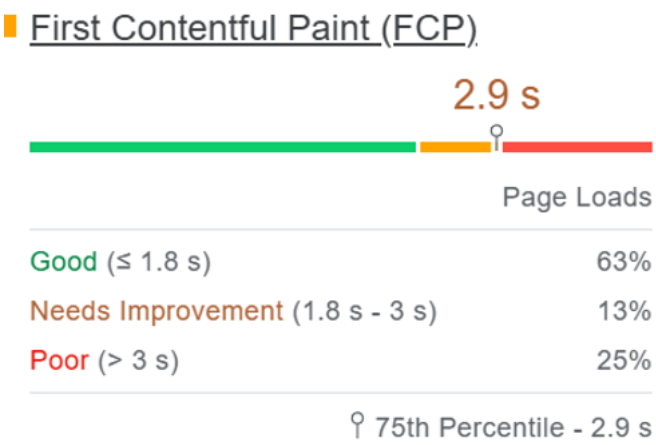
Both a responsive mobile website and a standalone app for iOS and Android are used by Shopee Malaysia. The brand makes mobile-first users a top priority because 59.93% of all site visitors are coming from the mobile web (SimilarWeb, 2026).

2.4.1 Mobile Page Load Times and SEO

A clear gap in technical performance exists for the mobile web interface. As per Figure , a 6.6-second Largest Contentful Paint (LCP) and a 2.9-second First Contentful Paint (FCP). These results are much slower than the 2.5-second limit required by Google's Core Web Vitals. Shopee puts a lot of effort into Mobile SEO to make up for these slow speeds. To keep product

listings appearing high in mobile SERPs, the site uses structured data schema and AMP (Accelerated Mobile Pages) principles. Even when a page takes too long to load fully, this optimization helps the brand keep a high level of organic visibility.





3.0 Data Collection

For data collection we focused on public user feedback about Shopee Malaysia. We used 'Apify' to scrape reviews from two sources. The first source was the Google Play Store, using the official Shopee app listing. The second source was the Shopee website itself Trustpilot review scraper agent to view the publicly available data.

We kept the scraping scope tight by only collecting content that is clearly related to the Shopee brand. The main selection rule was the brand name itself, which makes the data relevant to the study. We did not collect any private information and only kept what is already shown publicly.

Then we combined the scraped outputs into one raw file and saved as Shopee_Reviews.csv; which contains 2,793 reviews. This dataset includes the review date, star rating, review text, and user display name. This raw dataset was archived to support transparency and it is used as the starting point for cleaning and sentiment analysis.

4.0 Data Processing & Wrangling

We used Google Colab as our python environment because it is simple to set up and works well for Python based text analysis. Colab also helped us keep the process transparent since the code output and results can be reviewed and reproduced easily.

4.1 Loading Required Libraries:

First, we started by importing the libraries needed for the full workflow **(Figure: 10).**

```
# Install necessary libraries
!pip install deep-translator vaderSentiment

import pandas as pd
import numpy as np
import re
from deep_translator import GoogleTranslator
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import Markdown, display
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

Figure 10: *List of Libraries Used in Data Wrangling and Visualization*

4.2 Loading Dataset:

We loaded the raw file using pd.read_csv ('Shopee_Reviews.csv') **(Figure: 11).**

```
try:
    df = pd.read_csv('Shopee_Reviews.csv')
    print("✅ Data Loaded Successfully!")
except FileNotFoundError:
    print("❌ Error: 'Shopee_Reviews.csv' not found. Please upload it.")

```

```
✅ Data Loaded Successfully!
```

Figure 11: *Loading the Dataset for Analysis*

4.3 Duplicate Cleaning:

Dropped of duplicates through df.drop_duplicates(inplace=True) then verified through df.shape**.**

4.4 Removing Missing Values:

We handled missing values using two loops: Mean for numeric column and mode for text column. Later verified missing values using df.isnull().sum(). The output shows 0 missing values for Date, Rating, Review, and User **(Figure: 12)**.

	0
Date	0
Rating	0
Review	0
User	0

dtype: int64

Figure 12: *Dataset with no Missing Values*

4.5 Data Transformation & Reason

We translated reviews into English to support the sentiment tool later using GoogleTranslator(source='auto', target='en'). This allows automatic language detection and English output.

To ensure analysis of 'emoji only reviews', we built an emoji detector using a compiled regex called EMOJI_PATTERN. It includes ranges for emoticons, symbols, transport icons, flags, and other emoji blocks.

Then we defined a function called translate_review(text). If the text contains only emojis, it keeps the original input without translation. If it is normal text of different language, it tries to translate using translator.translate(text). If translation fails, it retains the original text as it was. Finally, the language translated outputs are stored in a new column named 'Translated_Reviews. This gave us a clean English version of the reviews, while still keeping emoji only reviews unchanged **(Figure: 13)**.

```python
# Translation (Preserving Emojis) ---
print("🔄 Starting Translation (This may take a few minutes)...")
translator = GoogleTranslator(source='auto', target='en')

# Regex to detect if a string contains only emojis
EMOJI_PATTERN = re.compile(
    "["  # Start character set
    "\U0001F600-\U0001F64F"  # emoticons
    "\U0001F300-\U0001F5FF"  # symbols & pictographs
    "\U0001F680-\U0001F6FF"  # transport & map symbols
    "\U0001F1E0-\U0001F1FF"  # flags (iOS)
    "\U00002702-\U000027B0"
    "\U000024C2-\U0001F251"
    "]+"
)

def translate_review(text):
    if not isinstance(text, str):
        return ""

    # Check if the text is composed ONLY of emojis
    if EMOJI_PATTERN.fullmatch(text.strip()):
        return text.strip() # Return original text if it's just emojis

    # Otherwise, proceed with translation
    try:
        return translator.translate(text)
    except:
        return text

df['Translated_Review'] = df['Review'].apply(translate_review)
print("✅ Translation Complete.")

df.head()
```

```
🔄 Starting Translation (This may take a few minutes)...
✅ Translation Complete.
```

	Date	Rating	Review	User	Translated_Review
0	01/02/2026	1	kedekut syiling emas bila digunakan untuk pemb...	Mohd Fariq	being stingy with gold coins when used to purc...
1	01/02/2026	5	I personally think that this is the best app I...	likchan622 likchan622	I personally think that this is the best app I...
2	01/02/2026	5	terbaik	Hisham Sabdin	the best
3	01/02/2026	5	Hope best services thanks	Raine WM	Hope best services thanks
4	01/02/2026	5	ok	Aliff Syazwan	ok

Figure 13: *Translation of Data for Analysis*

5.0 Manual Labelling for Model Accuracy Score

We saved the translated dataset and went through the reviews manually to label the sentiments as Positive, Neutral and Negative. Doing this we created a gold standard sentiment label against which the predictive model results will be compared for the accuracy.

6.0 Sentiment Analysis (Unsupervised Approach)

6.1 Setting Up the Rule Based Model (VADER)

We used a lexicon based approach using **VADER**. This method does not need any labelled training data. It works by using a built in sentiment dictionary and simple rules. We first initialized the analyser using SentimentIntensityAnalyzer(). **(Figure: 14)**

```
# Initialize VADER
analyzer = SentimentIntensityAnalyzer()
```

Figure 14: *Setting up VADER Analyser*

6.2 Getting the Sentiment Score from Text

For each review, we analysed a compound score using VADER **(Figure: 15)** which ranges from -1 to +1. A higher score means more positive sentiment and a lower score means more negative sentiment. We applied this on the Translated_Review column so that mixed Malay and English reviews can be handled in one flow.

```
    # 1. Run Standard VADER Analysis FIRST
    scores = analyzer.polarity_scores(text)
    compound = scores['compound']
```

Figure 15: *Sentiment Score Generation*

6.3 Converting Scores into 3 Sentiment Labels:

After scoring, we converted the compound score into three labels **(Figure: 16):**

- Positive (≥0.05),
- Neutral (-0.05 to < 0.05),
- Negative (≤ -0.05).

We stored the outputs in two new columns called **Sentiment_Score_Vader** and **Sentiment_Label_ Vader.**

```python
# Initialize VADER
analyzer = SentimentIntensityAnalyzer()


# Sentiment Analysis (3 Labels)
print("🔄 Running VADER Sentiment Analysis with 3 Labels...")

def get_sentiment_data(text):
    text = str(text)

    # 1. Run Standard VADER Analysis FIRST
    scores = analyzer.polarity_scores(text)
    compound = scores['compound']

    # Define 3 Sentiment Categories based on Compound Score
    if compound >= 0.05:
        label = 'Positive'
    elif compound <= -0.05:
        label = 'Negative'
    else:
        label = 'Neutral'

    return compound, label


# Apply function
df[['Sentiment_Score_Vader', 'Sentiment_Label_Vader']] = df['Translated_Review'].apply(
    lambda x: pd.Series(get_sentiment_data(x))
)

print("✅ VADER Sentiment Analysis Complete.")
```

```
🔄 Running VADER Sentiment Analysis with 3 Labels...
✅ VADER Sentiment Analysis Complete.
```

Figure 16: *Sentiment Categorisation*

6.4 Dominant Themes Using Topic Segmentation

To report dominant themes, we also grouped reviews into six topics. The topics were General Satisfaction, Transaction and Refund Services, Customer Support, Delivery Services, App Features and Livestream, and Technical Features. The final topic was saved as **Topic_Name**.

6.5 Summary of Sentiment Patterns

From 2,793 reviews, positive sentiment was higher overall. Positive was **1,552 reviews (55.6%)**, and Negative was **949 reviews (34.0%)**. Only **292 reviews (10.4%)** were sentimentally Neutral. Theme wise, the most reviews fell under **General Satisfaction (48.0%)**. **Transaction and Refund Services (22.3%)** and **Customer Support (12.6%)** tailed next to it **(Figure: 17)**.

Number of Reviews per Sentiment

```
Sentiment_Label_Vader
Positive    1552
Neutral      292
Negative     949
Name: count, dtype: int64
```

Percentage of Reviews per Sentiment

```
Sentiment_Label_Vader
Positive    55.567490
Neutral     10.454708
Negative    33.977802
Name: proportion, dtype: float64
```

Percentage of Reviews per Topic

```
Topic_Name
General Satisfaction             47.977086
Transaction & Refund Services    22.341568
Customer Support                 12.602936
Delivery Services                 8.557107
App Features & Livestream         6.158253
Technical Features                2.363050
Name: proportion, dtype: float64
```

Figure 17: *Summary of Sentiment Analysis of Non-labelled Data*

6.6 Visualisation for Non Labelled Sentiment

We created three visuals using **Sentiment_Label** and **Topic_Name**, to summarise the results. First, we plotted a pie chart **(Figure: 18)** for overall sentiment distribution. The chart uses the three sentiment labels i.e. Negative, Neutral, and Positive. The pie chart shows the percentage split for each sentiment class.

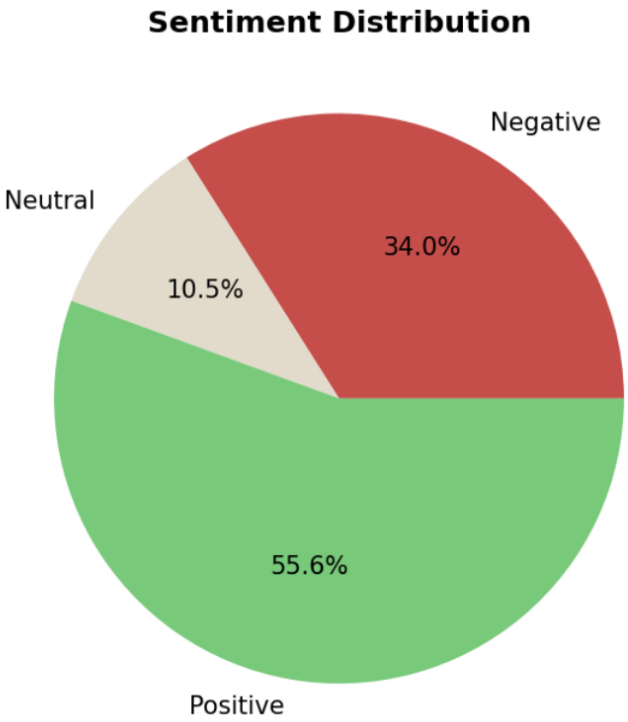


Figure 18: *Pie Chart of Sentiment Distribution*

Second, we plotted a topic frequency chart **(Figure: 19)**. This is a horizontal bar style count plot. It shows how many reviews fall under each topic. This chart helps us understand which areas are discussed by users the most.

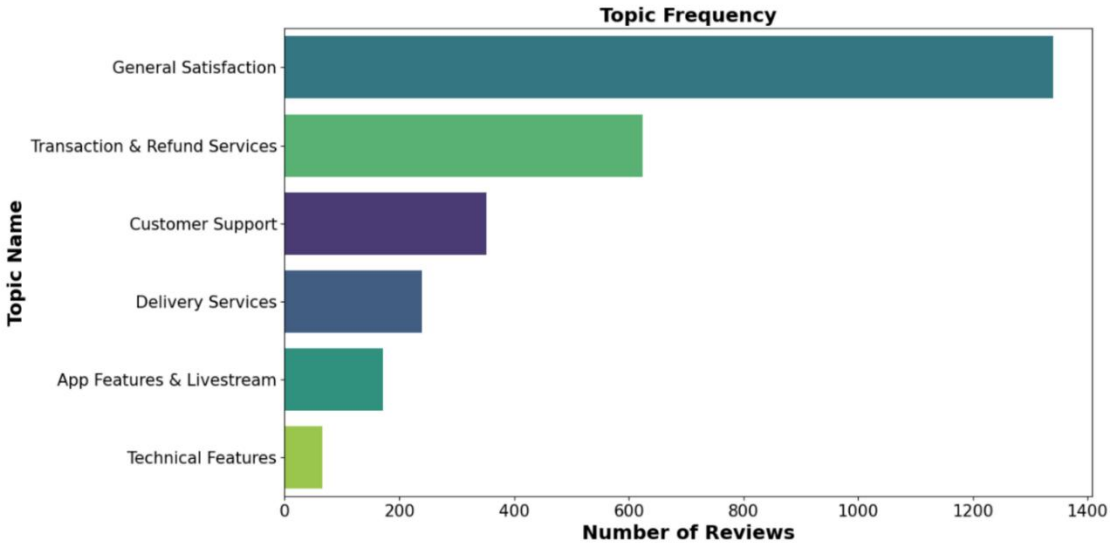


Figure 19: *Topic Frequency Bar Chart*

Third, we plotted a stacked bar chart for sentiment by topic **(Figure: 20)**. For this, we created a cross-tab table forced the sentiment columns into the it. After that, we plotted the stacked bars so each topic shows the full mix of sentiment labels. We also used the same colours so the sentiment colours match the pie chart. This chart helps us compare topics and see where negative sentiment is more concentrated.

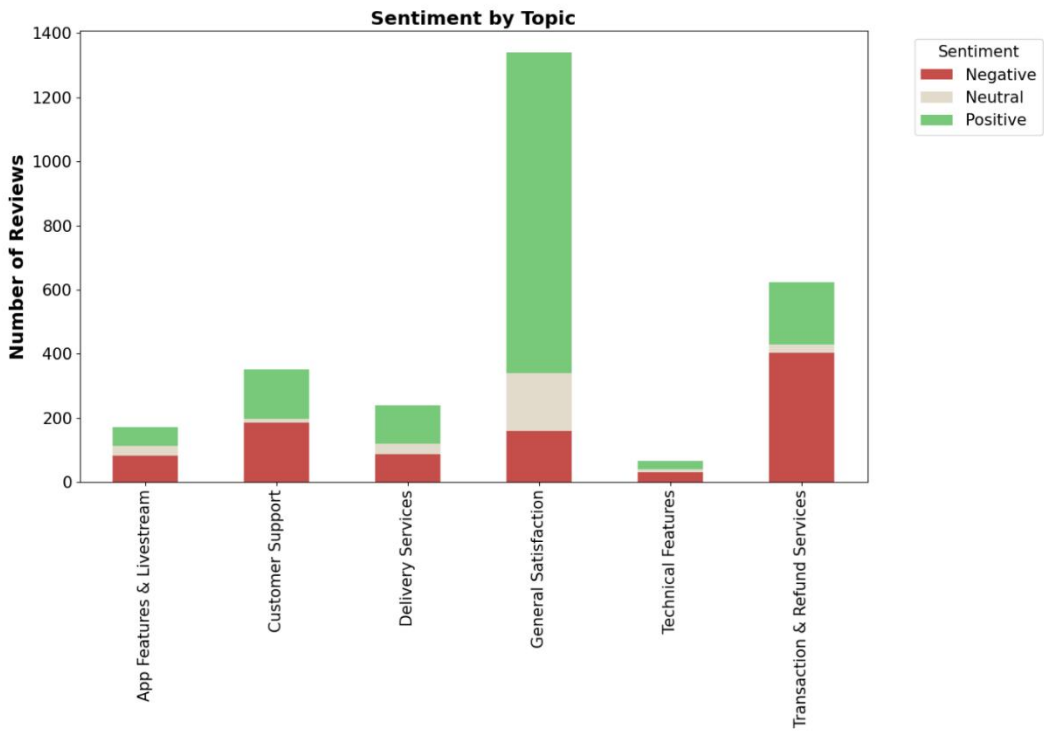


Figure 20: *Review Sentiment by Topic*

At the end of this section, we also printed the cross tab table in a clean format **(Figure: 21)**. This gives the exact counts behind the stacked bar chart. This is useful for checking the chart values.

```
DETAILED BREAKDOWN
Sentiment_Label_Vader          Negative  Neutral  Positive
Topic_Name
App Features & Livestream            82       30        60
Customer Support                    185       12       155
Delivery Services                    87       34       118
General Satisfaction                161      180       999
Technical Features                   31        9        26
Transaction & Refund Services       403       27       194
```

Figure 21: *Sentiment Distribution by Topic*

6.7 Unsupervised Results Validation (Model Accuracy)

Even though VADER is a rule-based method, we still checked how reliable it is. We compared VADER predictions with a manually labelled sentiment column in our dataset. The overall accuracy of the VADER sentiment model was 77.37%.

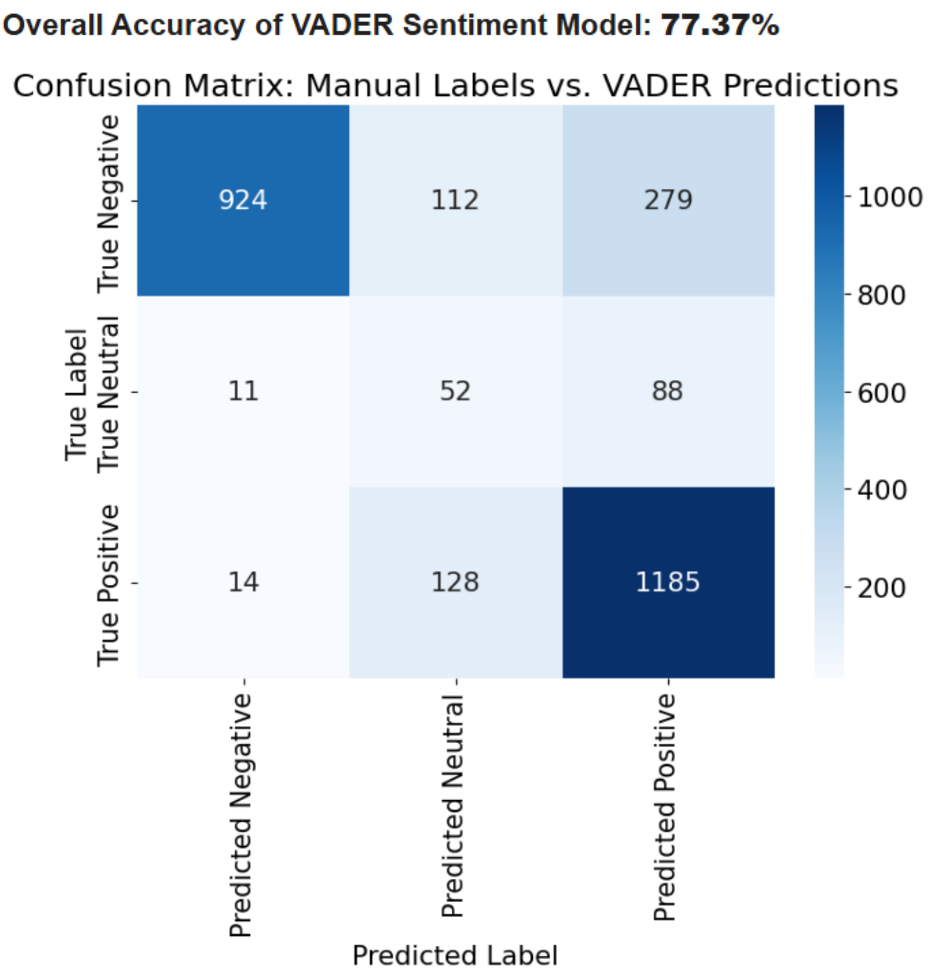


Figure 22: *Confusion Matrix for Unsupervised Learning*

The confusion matrix shows where VADER performs well and where it struggles **(Figure: 22).**

- For True Negative reviews, among 1315 total, VADER correctly predicted 924 as Negative, wrongly predicted 112 as Neutral and 279 as Positive. That is a **29.73%** deviation **(Figure: 23)**.
- For True Neutral reviews out of total 151 reviews, VADER correctly predicted only 52 as Neutral. Many Neutral reviews were pushed to Positive (88) or Negative (11). That is a **65.56%** deviation. This is the highest error rate among the three classes **(Figure: 23)**.
- For True Positive reviews among 1327, VADER correctly predicted 1185 as Positive, wrongly predicted 128 as Neutral and 14 as Negative. That is a **10.70%** deviation **(Figure: 23)**.

	Total Manual Labels	Correctly Predicted by VADER	Deviation (Misclassified)	Deviation Percentage
Negative	1315	924	391	29.73%
Neutral	151	52	99	65.56%
Positive	1327	1185	142	10.70%

Figure 23: *Deviations from Manual Labels*

The classification report **(Figure: 24)** supports this pattern. Negative predictions had very high precision (**0.974**) but lower recall (**0.703**). Positive predictions had strong recall (**0.893**) with good F1 (**0.823**). Neutral performance was weak. Precision was **0.178** and F1 was **0.235**. This is common in sentiment tasks because neutral text is harder to detect using lexicon rules.

	precision	recall	f1-score	support
Negative	0.973656	0.702662	0.816254	1315.00000
Neutral	0.178082	0.344371	0.234763	151.00000
Positive	0.763531	0.892992	0.823203	1327.00000
accuracy	0.773720	0.773720	0.773720	0.77372
macro avg	0.638423	0.646675	0.624740	2793.00000
weighted avg	0.830811	0.773720	0.788118	2793.00000

Figure 24: *Classification Report*

7.0 Supervised Approach

7.1 Dataset Information

Translated_Review (content to be analysed) and Manually_Labelled_Sentiment (ground truth for model training and evaluation) serve as the basis for sentiment analysis.

The dataset included the following columns: Date, Rating, Review, User, Translated_Review, Manually_Labelled_Sentiment, cleaned_text, sentiment_encoded, label

7.2 Data Preprocessing

Below are the steps involved in data preprocessing to ensure the models capture sentiment patterns effectively and improve classification accuracy:

1. **Ensure Uniformity:** Converting all characters into lower text
2. **Removing noises:** Links, @user and #tags were removed.
3. **Punctuation and White Space:** Fullstop, commas, space, and brackets were removed from text.
4. **Stopword:** Common English stopwords (e.g. “are”, 'the', 'is', 'and') were removed.

7.3 Sentiment Distribution

The dataset revealed a degree of class imbalance. Positive (47.5%) and Negative (47.1%) are almost equally represented as the most frequent sentiments, while Neutral (5.4%) sentiment accounts for only a small fraction of the total data **(Figure: 25)**.

Sentiment Label	Count
Positive	1327
Negative	1315
Neutral	151

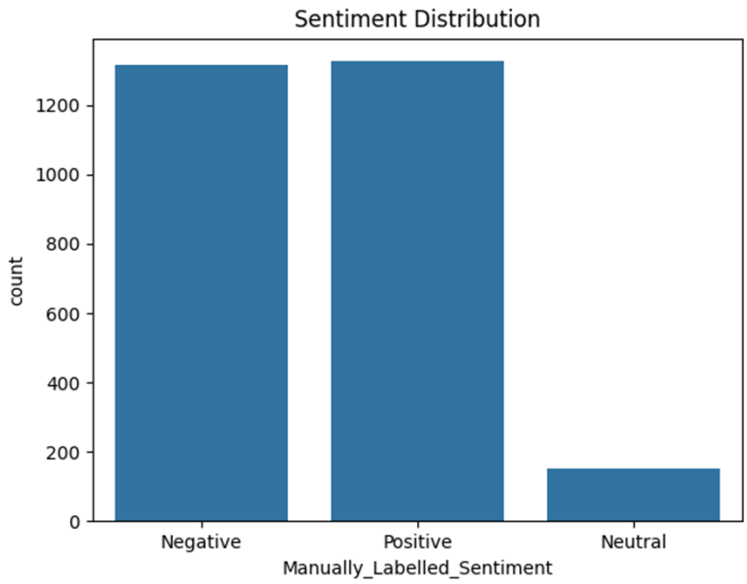


Figure 25: *Sentiment Distribution*

It is anticipated that limited Neutral sentiment samples will make it challenging for the model to generalize this category.

Top 20 Most Common Words (after preprocessing):

Word	Frequency
shopee	1613
seller	616
good	584
service	526
customer	463
app	434
refund	420
delivery	405
order	396
time	353
item	340
even	317
get	287
like	262
money	262
dont	258
shopping	250
product	247
days	241
use	237

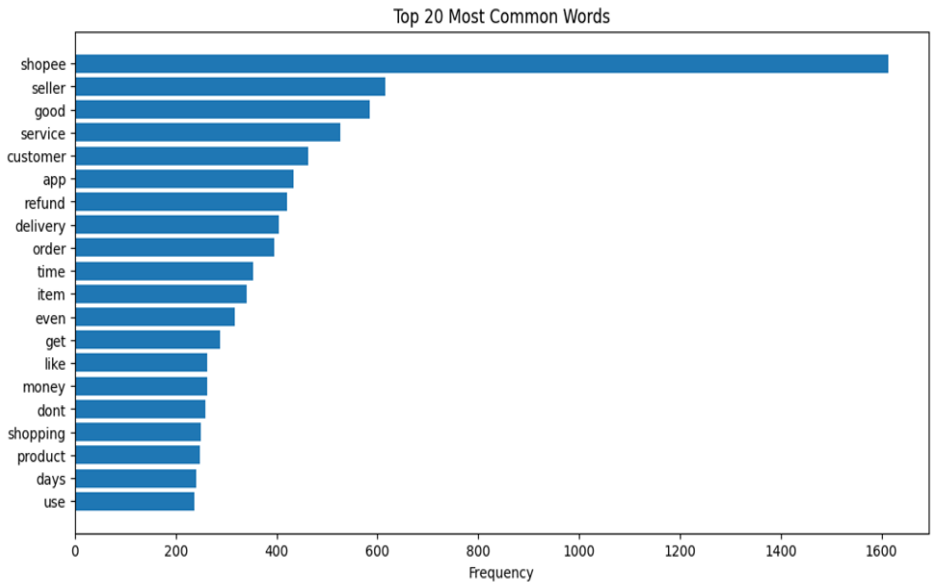


Figure 26: *Most Commonly Used Words*

The Top 20 Most Common Words reveal dominant themes that users discuss most in the review. As expected, the word "shopee" appeared most often, 1,613 times, given that the platform itself is the central subject of discussion (**Figure: 26)**. Overall, the topics that are most frequently discussed are platform, sellers, customer service, product, value and delivery processes.

7.4 Data Splitting and Vectorization

Each category of sentiment labels was assigned to a unique integer (0,1 and 2). Then we took the train-test split approach with the dataset. Training used 2,234 samples, which was 80% of the data. Testing used the remaining 20%, which was 559 samples. This split was done to make sure each sentiment class was fairly represented in both sets.

7.5 Supervised Learning Method

Different models required different approaches to handle text.

7.5.1 Logistic Regression

It applied TF-IDF to capture how frequently the term appears and how important it is across all reviews. It is efficient, easy to interpret and provides a strong baseline to classify text.

7.5.2 Naive Bayes

This used Count Vectorization which only focuses on raw word counts. It is known for its fast training and effectiveness with word frequency data.

7.5.3 DistilBERT (Deep Learning)

Unlike the traditional model, DistilBERT tokenised the text into contextual embeddings, which helped it understand meaning and subtle differences rather than just counting words.

7.6 Model Performance

7.6.1 Accuracy Comparison

The three models were evaluated using accuracy and classification reports.

4.1. Logistic Regression

- **Accuracy:** 0.8730 (87.30%)
- **Classification Report:**

Class	Precision	Recall	F1-Score	Support
Negative	0.90	0.92	0.91	263
Positive	0.95	0.86	0.90	266
Neutral	0.33	0.50	0.39	30
Macro Avg	**0.72**	**0.76**	**0.74**	**559**
Weighted Avg	**0.89**	**0.87**	**0.88**	**559**

We got a score of 87.3% of accuracy from Logistic Regression. The model is solid on Negative and Positive reviews, both with F1 scores around 0.90. But Neutral is where things fall apart. Precision of 0.33 means two out of three Neutral predictions were wrong, and recall shows it only caught about half of the actual Neutral cases.

4.2. Naive Bayes

- **Accuracy:** 0.8748 (87.48%)
- **Classification Report:**

Class	Precision	Recall	F1-Score	Support
Negative	0.90	0.92	0.91	263
Positive	0.86	0.93	0.90	266
Neutral	0.00	0.00	0.00	30
Macro Avg	**0.59**	**0.62**	**0.60**	**559**
Weighted Avg	**0.83**	**0.87**	**0.85**	**559**

Naive Bayes achieved a similar accuracy of 87.5%. Like Logistic Regression. But when look at the Neutral row, it's all zeros. The model didn't get a single Neutral case right. Everything was forced into either Negative or Positive. Naive Bayes depends on word frequency, and Neutral reviews don't have strong words that stand out. Which makes the model just guesses one of the other two classes

4.3. DistilBERT (Deep Learning)

- **Accuracy:** 0.9070 (90.70%)
- **Classification Report:**

Class	Precision	Recall	F1-Score	Support
Negative	0.91	0.96	0.93	263
Positive	0.92	0.92	0.92	266
Neutral	0.57	0.27	0.36	30
Macro Avg	**0.80**	**0.72**	**0.74**	**559**
Weighted Avg	**0.90**	**0.91**	**0.90**	**559**

In terms of results, DistilBERT came out on top with 90.7% accuracy. It was very strong on Negative and Positive, 0.93 and 0.92 respectively. Neutral was still the hardest class. Its F1-score was 0.36, which shows the model could at least recognize Neutral better than Naive Bayes.

In summary, the comparison shows that DistilBERT outperformed traditional models by 3% **(Figure:27)**. Neutral sentiment stayed the most difficult across all three models, due to class imbalance and ambiguous wording often found in Neutral Reviews.

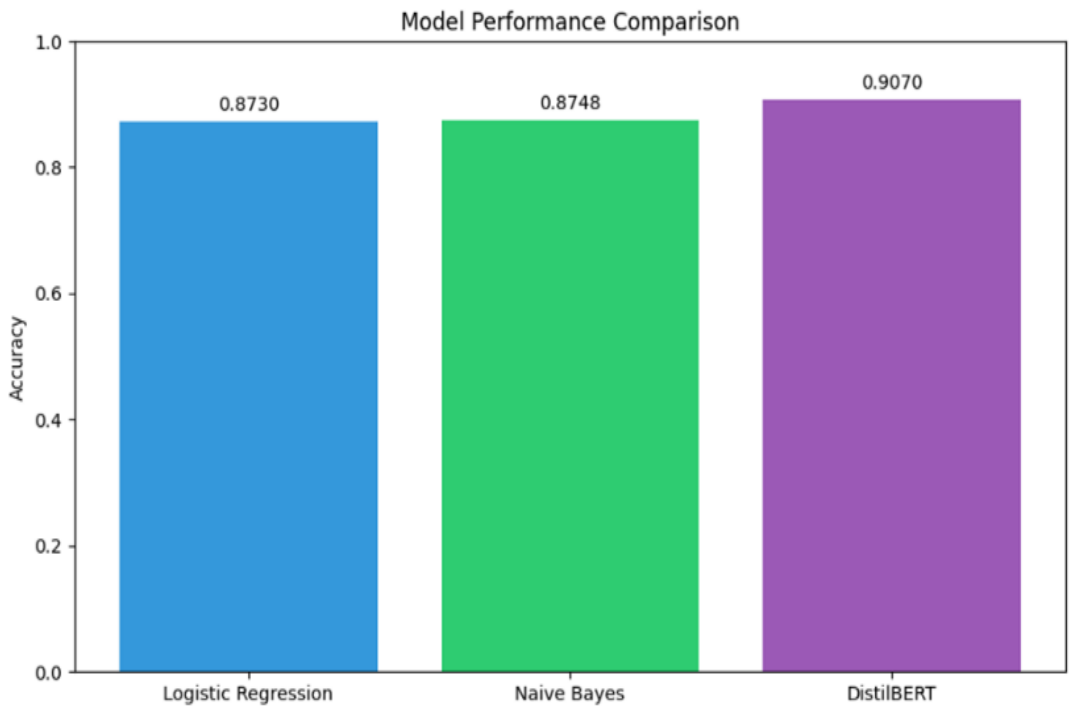
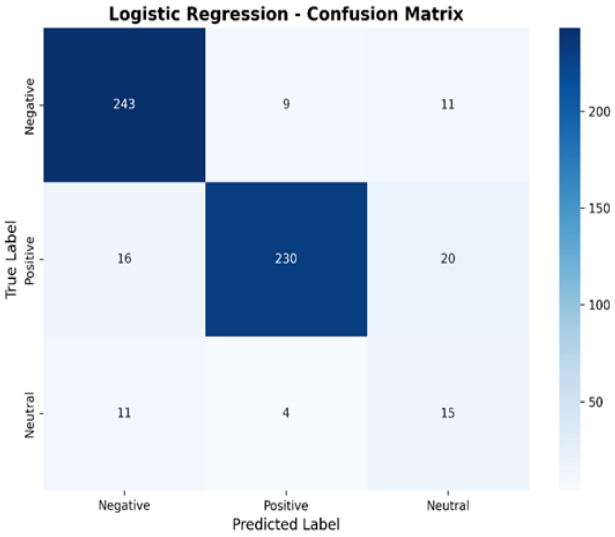


Figure 27: *Model Performance Comparison*

7.6.2 Confusion Matrix

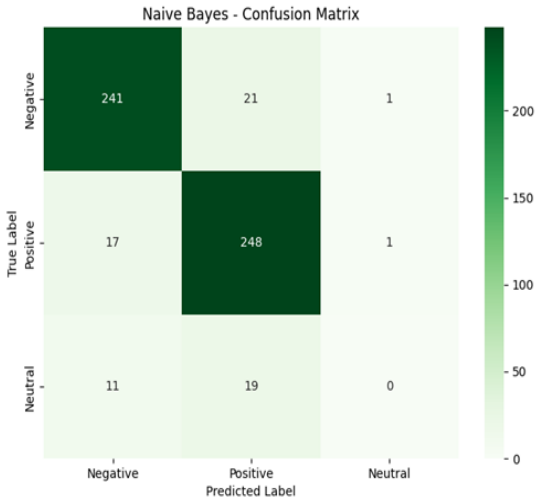
Next, we plotted three confusion matrices to compare the performance of the three classification models:

- Logistic Regression
- Naive Bayes
- DistilBERT.



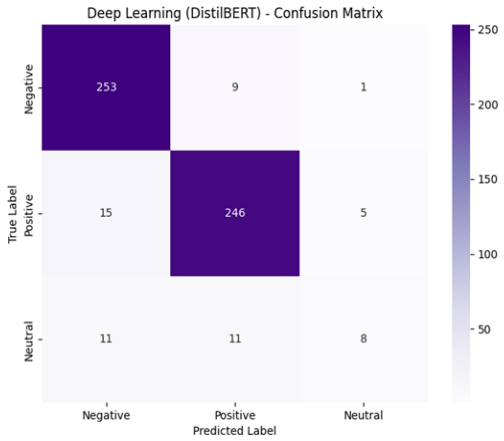
For Logistic Regression, the model correctly predicted 243 Negative sentiments, where original were 263 and 230 Positive sentiments out of 266. Neutral cases were much harder to predict, as there are only 15 out of 30 were predicted correctly **(Figure: 28)**.

Figure 28: *Logistic Regression Confusion Matrix*



Naive Bayes performed a little better. 241 out of 263 Negative cases were predicted correctly along with 248 correct out of 266 Positive cases. However, none of the Neutral reviews was predicted correctly (**Figure: 29)**.

Figure 29: *Naïve Bays Confusion Matrix*



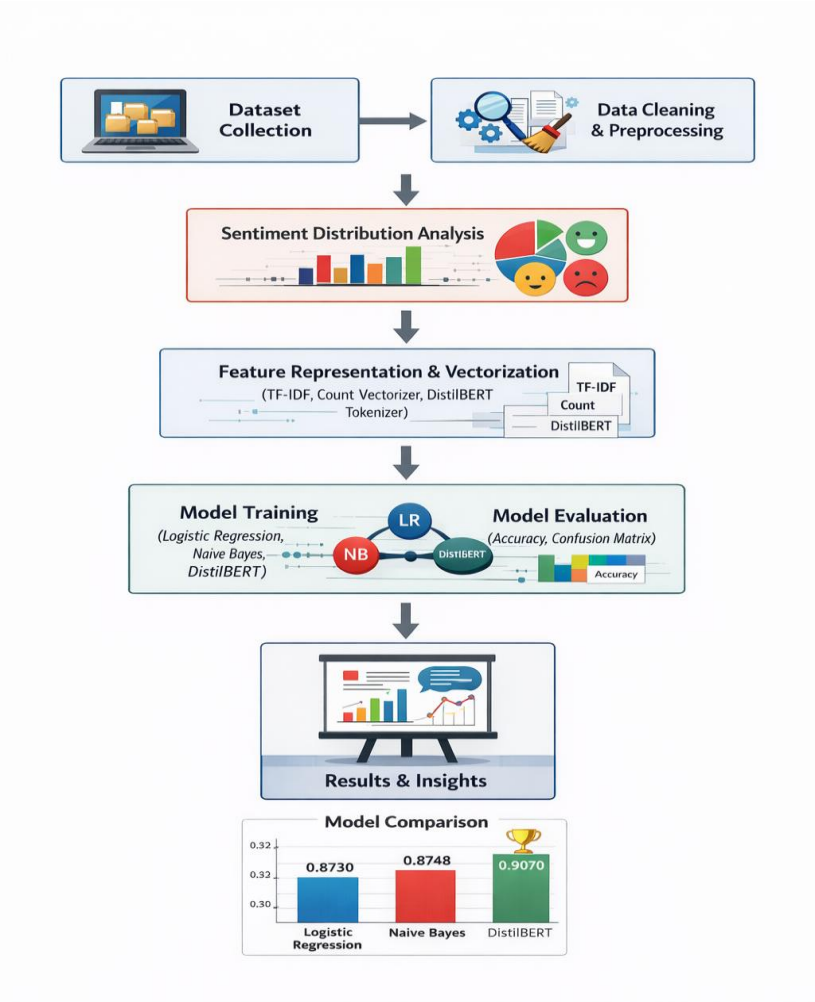
DistilBERT correctly predicted 253 out of 263 Negative reviews and 246 out of 266 Positive reviews. For Neutral, there are only 8 out of 30 were predicted correctly (**Figure: 30)**.

Figure 30: *DistilBERT Confision Matrix*

7.7 Model Comparison and Conclusion

DistilBERT shows the strongest overall accuracy at 90.7%. It does especially well on Negative and Positive sentiment. Interestingly, Logistic Regression performs better on Neutral, with an F1 score of 0.39. On the other hand, Naive Bayes had a decent overall accuracy of 87.5%. But it completely failed on Neutral, scoring zero across all the models.

7.8 Labelled Sentiment Analysis Workflow



The workflow diagram outlines the key stages of the sentiment analysis project. The process begins with dataset collection through web scraping. Followed by data cleaning and preprocessing to remove duplicates and reduce noise. Next, sentiment distribution analysis is performed to examine the balance between Positive, Negative and Neutral reviews, with Neutral sentiment showing noticeable imbalance. After this, the data is split into training (80%) and testing 20%. Vectorisation is applied using TF-IDF, Count Vectorizer, and DistilBERT. Three models: Logistic Regression, Naive Bayes, and DistilBERT are implemented to train

with these data. Finally, the model's performance was evaluated from accuracy scores and confusion matrix. Among the other two models, DistillBERT are the one that achieved the highest accuracy score of 90.70%.

8.0 Comparison

In this section, we compare the performance of the Unsupervised (VADER) model against the Supervised (Machine Learning) models. This helps us understand which method is most reliable for tracking of Shoppe's brand health.

- Deep Learning vs. Lexicon Rules: The supervised DistilBERT model is the winner with a 90.70% accuracy rate. However, the unsupervised VADER model achieved only 77.37% and this shows that while VADER is good for quick, automated checks, it lacks the capability to understand the complex, mixed-language reviews (Malay-English) that Shopee users often write.
- The 'Neutral' Sentiment Struggle: Both models found Neutral sentiment difficult to detect. VADER had a massive 65.56% error rate for neutral reviews. Even the high performing DistilBERT struggled here because neutral language is often subtle and lacks strong "emotion words."
- Sentiment Distribution Gap: There is seen a difference in how the models have taken the data. VADER predicted 55.6% positive sentiment. However, when we look at the manual labels used for supervised learning, the count for positive (1,327) and negative (1,315) was almost equal. This suggests that a basic automated tool like VADER may be "too idealistic" and miss nearly 13% of critical negative feedback.

Model Type	Model Name	Accuracy	Reliability
Supervised	DistilBERT	90.70%	Highest
Supervised	Logistic Regression	87.48%	High
Supervised	Naïve Bayes	87.30%	High
Unsupervised	VADER	77.37%	Moderate

Table 2: *Accuracy Comparison of Models*

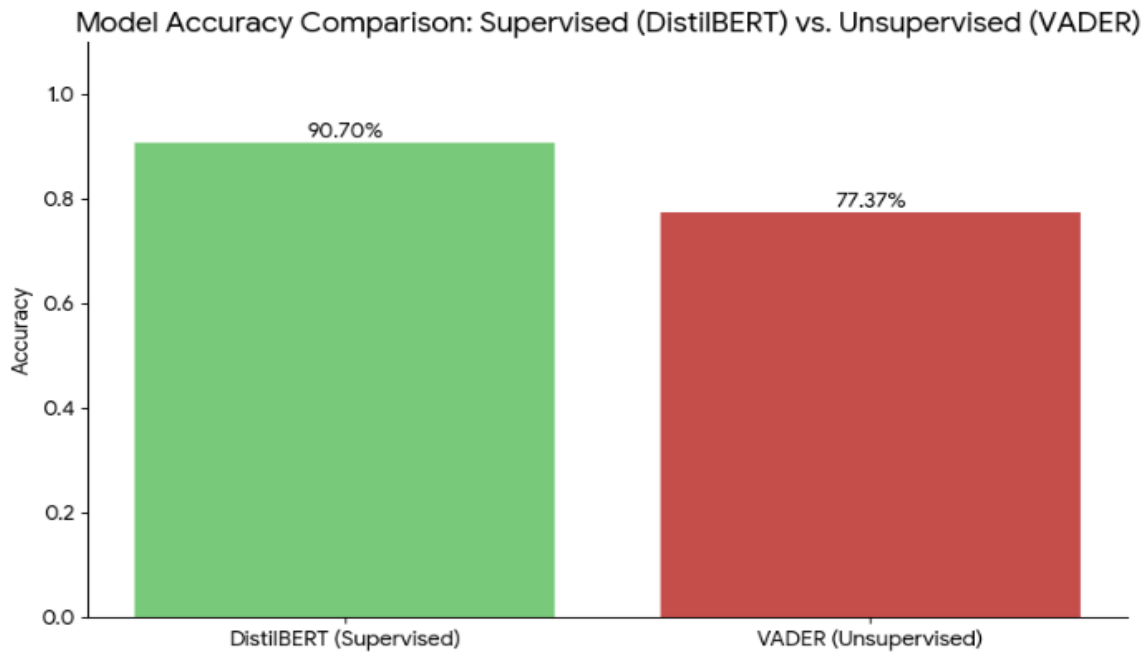


Figure 31: *Model Accuracy Comparison (Supervised Vs Unsupervised)*

8.1 Strengths and Weaknesses

Model	Strength	Weakness
VADER (Unsupervised)	Strength: It does not require any labelled training data to start. It is fast and efficient for getting a general overview of a large dataset immediately.	It is extremely poor at detecting the 'Neutral' sentiment, with a 65.56% error rate. It struggles with mixed-language (Malay-English) reviews and understand sarcasm.
Naive Bayes (Supervised)	It is very fast in training speed and highly effective when working with word-frequency data.	It assumes that every word in a review is independent of others, which means it can miss the deeper 'context' or meaning behind phrases.
Logistic Regression (Supervised)	It provides a strong baseline and is interpretable, which is easy to see how specific words influence the sentiment score.	Although dependable, it lacks the refinement of deep learning to understand complex sentence structures.
DistilBERT (Supervised Deep Learning)	The highest accuracy at 90.70%. Because it uses "contextual embeddings," it understands the relationship between words in a sentence, capturing nuance that other models miss.	It requires more computational power and time to train compared to traditional models.

Table 3: *Strengths and Weaknesses of Analytical Models Used*

9.0 Discussion and Future Recommendation

After interpreting the data and user reviews, it is clear that while Shopee is a market leader, it is currently facing an issue of "user experience (UX) and logistic trust."

9.1 Key Discussion points

- The 'TikTok-ification' Backlash: A major theme in the negative reviews is the app's new layout. Users are frustrated that the app "defaults to Live & Video" on startup instead of the Home page. Many has described this as "annoying" and "data-hungry", with loud audio that starts automatically.

- Logistic Failures (SPX Express): While Shopee offers "Free Shipping," the quality of its internal courier, SPX Express, is a primary source of angry. Users reported items

stuck at hubs (like the OUG hub) for weeks and riders filing “fake” delivery attempts when they never actually showed up.

- The ‘SI Support’ Wall: Users feel trapped behind unhelpful chatbots. Reviews frequently mention that agents provide “automatic answers” and “pre-written replies,” instead of exact solution for their query. Also, that the “refund process” is slow, often taking weeks or months to resolve.

9.2 Supervised Approach interpretation

The 90.7% accuracy of the DistilBERT model shows that deep learning is much more effective than traditional methods for understanding local e-commerce feedback. This higher accuracy allows Shoppee to actually move past text review and understand user intent behind the user review, and addressing those sentiments accordingly.

- ***The Emotional Polarization of Users****:* While the model performed well, the high error rate in neutral sentiment suggests that Shopee users rarely express middle-ground opinions. Most feedback is highly polarized, with users being either very vocal about their satisfaction with pricing or extremely frustrated by service gaps. As a result, Shopee misses the opportunity to collect constructive feedback for products/services that are acceptable but still require improvements and can help improve sales of those categories.
- ***Processing Mixed-Language Feedback****:* DistilBERT performed better than traditional models because it analyzes how words relate to each other in a sentence. This ability was essential because Malaysian customers often use a mix of Malay and English or add local words like "lah". Basic word-counting models often fail to classify these patterns correctly because they do not understand the surrounding context.
- ***Risk Detection:*** The model’s 0.93 F1-score for negative sentiment provides a dependable way to find serious logistics failures or technical bugs. By accurately flagging intense frustration regarding SPX Express or refund delays, Shopee can prioritize these "at-risk" cases before customers switch to competitors.

9.3 Unsupervised Approach Interpretation

The unsupervised analysis gives a broad view of the brand's reputation by automatically sorting the 2,793 reviews into specific themes. While this method is less precise than deep learning models, it points out where the highest volume of customer complaints is located.

- ***Negative Sentiment in Refund Services:*** The model sorted reviews into six categories and found that "Transaction and Refund Services" accounts for 22.3% of the total reviews. The high amount of negative feedback in this specific area proves that the refund process is currently a major risk to customer retention.
- ***Overestimation of Positive Feedback*:** VADER predicted a 55.6% positive sentiment rate, which is much higher than the manual results. This happens because the rule-based tool is often too idealistic and misses sarcasm or ironic negative reviews. For Shopee, this means relying only on basic tools could lead the company to ignore roughly 13% of critical complaints, which could have been used to improve their services.
- ***Lack of Detail in Categorized Feedback:*** Since the model assigned 48% of the reviews to "General Satisfaction" whenever a keyword did not match, nearly half of the data lacks specific detail. This creates a blind spot where growing technical issues or niche complaints are oversimplified and could be missed by the business.

9.4 Future Strategic Recommendations

1. Introduce a “Home Page” Toggle: Shopee should allow users to choose their default landing page. Forcing shoppers into livestreams is driving loyal customers towards competitors like Lazada or TikTok Shop, while some users feel are now easier to use.
2. Courier Accountability System: To fix “fake delivery” issue, Shopee should implement a mandatory GPS verified photo for all “unsuccessful” attempts. The data shows users are losing trust in the “On-Time Guarantee” because it feels like a marketing trick rather than a real promise.
3. Modernize UI Features: Surprisingly, the dataset contains many requests for a native Dark Mode in 2026. This is a simple technical update that would significantly improve “General Satisfaction,” which is currently the largest topic of discussion (48%).

4. Tiered Customer Support: Shopee needs to move away from 100% AI-driven support for high-value disputes. For refunds involving large amounts (e.g., over RM100), users should be automatically routed to a 'human agent' to prevent the issues where they can't get help or delete their account.

10. Conclusion

This conclusion summarizes the findings from our complete analysis of 2,793 reviews regarding Shopee Malaysia. By combining web metrics, data scraping and advanced sentiment modelling, we have gained a clear understanding of how users feel about the platform and how Shopee can improve its services.

The comparison of our models proved that 'supervised machine learning' is the most effective way to track user feelings. The DistilBERT model is the winner with 90.70% accuracy rate. This model is much better at understanding the way people actually talk compared to the VADER model which often got confused by neutral or mixed-language reviews. A key finding is all the models found 'neutral' reviews the hardest to categorize because these reviews often contain facts rather than emotional words. Our sentiment analysis showed a platform divided between love and frustration, with 1,327 positive and 1,315 negative reviews. On the positive side, users continue to praise Shopee for its wide product selection, low price, more of vouchers. However, the data shows that this loyalty is being tested by new app features that users find annoying. The most important complaints focused on the app's layout, specifically the automatic launching of Live & Video context. Users also expressed deep angry towards SPX Express logistics, citing as fake delivery attempts and items stuck at delivery for weeks. Furthermore, the AI chatbots for customer support has left users feeling ignored.

To maintain its dominant position, Shopee must shift its focus from engagement numbers to user comfort and trust. Introducing a simple toggle to disable auto-play videos and adding a native dart mode would solve the common complaints. Most importantly, Shopee needs to improve the accountability of its internal courier and provide easier access to human support for complicated refund cases.

Note: The accuracy of the DistilBERT model (90%+) proves that Shopee has the data capability to fix these issues. They are not 'missing' the feedback; they are simply not acting on it quickly enough.

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Appendix

- **Raw Dataset Link: Shopee_Reviews.csv**
- **Translated and Manually labelled Sentiment Dataset Link: Shopee_Reviews_Cleaned and Manually Labelled.csv**